

Differentiated Reliability-Based Regional Distribution Network Planning With Probability Characteristics of PV and Load

Ziyi Liu , Hong Liu , Member, IEEE, Zhengyang Xu , Ranfeng Li, Yue Guo, and Yongmei Du

Abstract—Increasing reliability demand and massive uncertain distributed PV bring new challenges to distribution network development. In this paper, a regional distribution network planning framework is proposed to consider differentiated reliability and probability characteristics of PV and load. Firstly, a robust optimization method for candidate feeder number from substations is introduced by leveraging flexible load rate. Secondly, the rotating center line algorithm is applied in load allocation optimization method. Finally, in view of nonlinear coupling relationship among the number of feeders, reliability cost, and investment, a regional distribution network planning model is formulated. Comparing to the method with fixed feeder number, numerical results demonstrate that the proposed method can divide the power supply range of feeders flexibly, while obtaining planning scheme with the balance of reliability and economy.

Index Terms—Balance of reliability and economy, flexible load rate, regional distribution network planning.

NOMENCLATURE

Indices and Sets

t	Index for time intervals.
m, n	Index for feeder and load.
g	Index for load rate group.
$\Omega_{L,s}^g$	Set of all loads in group g and central line set s .
Ω_h^{str}	Set of endpoints of all pending street segments in group h .
Ω_h^V	Set of all nodes in group h .
Ω_h^{Sub}	Set of all substation nodes in group h .

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Ziyi Liu, Hong Liu, and Zhengyang Xu are with the School of Electrical and Information Engineering, Tianjin University, Tianjin 300072, China (e-mail: liuziyicumt@163.com; liuhong@tju.edu.cn; xuzhengyang@tju.edu.cn).

Ranfeng Li is with the Henan Zhongyuan Energy Construction Engineering Company Ltd, Henan 450003, China (e-mail: 1245756845@qq.com).

Yue Guo and Yongmei Du are with the State Grid Economic and Technological Research Institute, Beijing 102209, China (e-mail: chinaguoyue@chinasperi.sgcc.com.cn; duuyongmei@163.com).

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Ω_h^{Tie}	Set of all tie switch nodes to be selected in group h .
$\Omega_i^{\text{child}}, \Omega_i^{\text{parent}}$	Set of child/parent nodes for node i .
G_h^n	Set of streets surrounding the load n .
Ω_h^{load}	Set of all loads in group h .
Parameters	
$\eta_g \%$	Maximum load rate in group g .
S_{line}	Maximum capacity of feeder.
$N_{\text{PV}}, N_L, N_{\text{DR}}$	Number of PV, conventional load and shiftable load participating in demand response respectively.
$N_{\text{PV},m}, N_{L,m}, N_{\text{DR},m}$	Number of PV, conventional load and load participating in demand response in feeder m respectively.
Γ_t	Uncertainty parameter.
α	Maximum proportion of shiftable load to the overall load.
β_1, β_2	Weighting factors.
N_{com}	Number of combined feeder blocks between different substations.
b_i^{bor}	Degree of overlap between the two feeder blocks.
f	Rate of discount.
r	Service lifespan of feeder.
$c_{\text{trk}}, c_{\text{brh}}$	Construction cost per unit length of main/branch feeder.
F_n	Reliability compensation coefficient.
L_n	Load value of load point n .
N_h^{Sub}	Number of substations in the constructed feeder in combination group h .
S_{Tie}	Capacity margin of connecting feeder.
W_{SRT}	Total amount of load that need to be switched.
$D_{n,ij}$	Distance from load node n to street segment ij .
N_i	Total number of consumers.
$ASAI_{\text{need}}$	Lowest bound of ASAI.

Variables

$P_{L,i}^t$	Conventional load demand power of i at time t .
$P_{\text{PV},i}^t$	Output power of PV unit i at time t .
$P_{\text{DR},i}^t$	Load demand response power of i at time t .
$P_{\text{line},m}^t$	Calculated transmission power of the feeder m .

$P_{PV,pre,i}^t$	Predicted benchmark value of photovoltaic Unit/load i at time t .
$P_{L,pre,i}^t$	Weighted/actual distance from load point n to central line m in group g .
d_{mn}^g, d'_{mn}	Weighted factor representing difference between load balance/peak and valley of line m in group g .
$\sigma_{lb,m}^g, \sigma_{pvr,mn}^g$	Angle of central line m in group g .
α_m^g	Number of feeders in group g .
N_{line}^g	Maximum load of all feeders / feeder m in load
$L_{max,s}^g$	rate group g and central line set s .
$L_{max,m,s}^g$	Degree of overlap between the two feeder blocks.
b_i^{bor}	Total cost of of each pair of interconnected feeders in group h .
C_h	Investment and operation cost for Main / branch feeders of single ring network of feeder m in group h .
$C_{h,m}^{tr}, C_{h,m}^{br}$	Annual power outage cost of feeder m in group h .
$C_{h,m}^{re}$	Length of feeder constructed between i and j / load n and main line.
len_{ij}, len_n	Power outage time of load point n after the fault of feeder section ij .
$t_{ij,n}$	Virtual power flow from node i to node j .
F_{ij}	Continuous variable indicating the order relationship of each node in the main trunk line.
q_i	Binary variable indicating whether construct a line between i and j .
χ_{ij}	Binary variable indicating whether streets around the load are constructed.
$z_{n,ij}$	Binary variable indicating whether the distance between node v and n is minimum value of load n .
$\phi_{n,v}$	Binary variable indicating whether the node v is included in constructed feeders.
κ_v	Binary variable indicating whether the load n can be transferred.
$s_{ij,n}$	Binary variable indicating whether the load n can be restored until the fault is repaired.
$\tau_{ij,n}$	Binary variables indicating upstream /mid-stream/downstream segment for branch ij .
$\delta_{ij}^U, \delta_{ij}^M, \delta_{ij}^D$	Binary variable indicating whether the load can be transferred.
v_{st}	

I. INTRODUCTION

THE output power of distributed generation is significantly influenced by natural conditions, such as sunlight which exhibits considerable intermittency and fluctuation. Moreover, the stochastic behavior of consumers leads to uncertainty in electricity demand, posing challenges to distribution network. In practical engineering applications, the primary objective of distribution network planning is to enhance the operational efficiency and economic performance of the system while ensuring

reliability. However, a highly reliable distribution network requires substantial investment and construction costs, which may compromise the economic feasibility of the planning scheme. Consequently, it is essential to develop a distribution network planning methodology that balances reliability and investment [1], [2], [3].

A. Literature Review

Current safety criteria adopted in distribution network planning can be broadly categorized into three types: traditional rigid N-1 constraints, reliability constraints, and reliability costs. Rigid safety constraints, as discussed in [4], [5], [6], [7], limit peak load to ensure it does not exceed line capacity during normal state. References [8], [9] impose the requirement that the system must be capable of transferring the load of a tie line before the fault recovery, which is defined as the rigid N-1 constraint. However, the rigid N-1 constraint often leads to relatively low usage of line capacity, a large power supply margin and excessive equipment investment.

Reference [9] incorporates the reliability index expected energy not supplied (EENS) as a constraint in distribution network planning integrated with DG. Moreover, reference [10] addresses the limitation of customer service unreliability at the load point. In [11], both consumer and system reliability indices are constrained to remain above a certain threshold, such as Customer Interruption Frequency (CIF), Customer Interruption Duration (CID), System Average Interruption Frequency Index (SAIFI), System Average Interruption Duration Index (SAIDI) and EENS.

In [1], [12], [13], reliability costs are computed based on EENS and integrated into the overall objective function in planning stage. References [14], [15] provide detailed methodologies for calculating EENS, SAIFI, and SAIDI in the context of Mixed-Integer Linear Programming (MILP) problems. Similarly, these reliability indices are converted into economic costs and incorporated into the objective function. In [16], [17], reliability costs are treated as a sub-problem of the planning process, where an independent objective function is formulated to minimize the economic losses associated with the loss of expected energy due to insufficient resources. Remarkably, the investment in feeders is directly proportional to their number, while the level of reliability is inversely proportional to the number of feeders. The aforementioned studies assume a fixed number of substation feeders, which limits the ability to comprehensively address both reliability and economic factors in the planning scheme. Although reference [8] explores the use of a variable number of feeders and geographical characteristics, it still relies on the rigid N-1 criterion, leading to excessive equipment investment. Consequently, the challenge of balancing the relationship between the number of substation feeders and the level of reliability remains unresolved.

In addition, uncertainty is a critical factor in many network planning studies. The modeling approaches commonly used to address uncertainty include stochastic optimization, robust optimization, distributed robust optimization, and information gap decision theory [18], [19], [20], [21], [22]. However, the impact

TABLE I
SUMMARIZE OF COMPARISON WITH PREVIOUS WORK

Ref	Scene	Number of feeders	Security criterion	Uncertain DG&load	Economy & reliability coupling
[1]	G	D	cost	✓	
[4]	N	F		✓	
[5]	N	F		✓	
[6]	G	F			
[7]	N	F		✓	
[8]	N	F	N-1		
[9]	G	D	N-1		
[10]	G	F	Str.		
[11]	N	F	Str.		
[12]	N	F	cost	✓	
[13]	N	F	cost	✓	
[14]	N	F	cost		
[15]	N	F	cost		
[16]	N	F	cost	✓	
[17]	N	F	cost		
The proposed method	G	D	Str&cost	✓	✓

Notes:

N: pre-selected nodes	G: geographical features
F: fixed range	D: decision variables
Cost: reliability cost	Str: reliability constraint

that uncertainty characteristics of PV and load bring on the number of feeders has not been thoroughly investigated during regional planning. There is an obvious knowledge gap in how to manage changes in the number of feeders from substations caused by the uncertainties in PV generation and load demand within the distribution network.

B. Research Gaps and Contributions

According to the above literature, it can be found that the total number of feeders from substations is not typically considered as a decision variable during the planning stage, regardless of the safety criteria employed, whether it be N-1 constraints, reliability constraints, or reliability costs. Consequently, it is difficult to develop an optimal planning scheme for regional distribution network. Specifically, there are complex coupling relationships among investment, reliability level and total number of all feeders from substations. Given this, it is essential to consider these factors comprehensively to achieve an optimal distribution network planning scheme. Moreover, the uncertainty characteristics of PV and load are also crucial factors influencing the number of feeders.

Focusing on the aforementioned research gaps, a regional distribution network planning framework is proposed considering differentiated reliability demand and probability characteristics of PV and load. Firstly, the robust range of net load within substations' service area is calculated by considering probability characteristics of PV and load. The upper and lower bounds of relaxed load rate are determined by the safety criteria. Meanwhile, integrating these bounds with net load values, the interval for the total number of feeders from substations can be obtained.

Then, in order to balance the net load and minimize the peak valley difference between feeders, a load partitioning method is proposed based on rotating center line (RCL) algorithm. Finally, aiming at finding the balance between reliability and economy, a planning optimization method is proposed for each feeder's service area. Table I summarizes the main differences between this work and the state-of-the-art in distribution network planning [1], [3], [4], [5], [6], [7], [8], [9], [10], [11], [12], [13], [14], [15], [16], [17].

The main contributions of this paper are as follows:

- 1) A robust optimization method is proposed for determining the total number of all feeders from substations, considering flexible load rate constraints and probability characteristics of PV and load.
- 2) An integrated optimization method is developed to address the nonlinear coupling relationships among the total number of feeders from substations, reliability costs, and investment costs, facilitating power supply scope division and planning in regional distribution networks.

The remainder of this article is organized as follows:

Section II presents for regional distribution network planning. A robust optimization to determine the interval of the total number of feeders from substations is introduced in Section III. Load partitioning based on rotating center line algorithm is conducted in Section IV. The network planning model considering both economic and reliability aspects is proposed in Section V. The solution process is described in Section VI. Finally, Section VII concludes the paper.

II. FRAMEWORK OF REGIONAL DISTRIBUTION NETWORK PLANNING

The objective of regional distribution network planning is to obtain an optimal scheme that considers reliability and economy comprehensively. The framework for this planning process incorporates several critical elements, including the probabilistic characteristics of uncertain PV and load, the optimization of the feeder count interval from substations, and network planning strategies.

A. Robust Optimization for the Interval of Feeder Numbers From Substations

Initially, the robust interval of net load within the power supply area of the substation is calculated, taking into account the probabilistic characteristics of PV and load. Then, according to the rigid N-1 criterion and the line capacity constraint, the upper and lower bounds of flexible load rate are determined. Finally, combined with the net load value in substation power supply range and relaxed load rate, the feasible interval for the total number of feeders from substations is derived.

The objective of this sub-problem is to calculate and identify all flexible load rate intervals and the corresponding sets of feeder numbers for the substations. This approach enables the enumeration of load rates and feeder numbers in subsequent optimization stages, where these values can be treated as fixed, thus simplifying the overall problem.

B. Load Partition Based on Rotating Center Line Algorithm

Firstly, all load points within substations are divided using rotation center line (RCL) algorithm. Optimal partition schemes are then selected based on a weighted sum of two key indices: load balancing and the minimization of peak-valley differences.

The purpose of this sub-problem is to allocate loads based on a predefined load rate and number of feeders, according to specific partitioning criteria. This allocation determines the loads each feeder will carry, effectively dividing the loads within a region into multiple feeder areas. Consequently, this division simplifies the routing problem in subsequent stage.

C. Network Planning Considering Economy and Reliability

Firstly, the connectivity between feeder blocks is analyzed to establish the interconnection relationships between feeders. Then, the main lines, branch lines, and segmented switches are optimized with the goal of achieving both optimal reliability and economic efficiency for each feeder. During the wiring stage, both the length of main lines and trunk lines are minimized. Additionally, the placement of sectionalizing switches on each feeder is optimized to distribute the load across segments as evenly as possible. Finally, a linear analytical expression is formulated to calculate the outage time of each load. Moreover, the reliability index EENS is converted into an economic loss using outage penalties associated with different loads in the objective function. Similarly, the ASAI index is constrained to meet the minimum reliability level required by each load.

The objective of this sub-problem is to simplify a complex regional routing problem by breaking it down into routing problems for each group of single tie-lines. Feeders are grouped into single tie-lines based on their degree of adjacency, and then routing is optimized for each group. By calculating the optimal reliability and economic efficiency for each group of single tie-lines, the overall optimal solution is achieved cumulatively.

Based on the process outlined above, the most reliable and economically efficient planning scheme is selected from all potential options. To facilitate this, a technical framework has been constructed, as illustrated in Fig. 1.

III. ROBUST OPTIMIZATION FOR THE INTERVAL OF TOTAL NUMBER OF ALL FEEDERS FROM SUBSTATIONS

Multiple feasible maximum load rates can be obtained by relaxing constraints based on rigid N-1 criterion. In this study, the N-1 criterion refers to a specific form of load rate constraint, where the maximum load rate of a line must not exceed 50%. This stringent constraint ensures that, after a fault, the tie line can fully transfer the load. Building upon this, we relax the constraint by setting 50% as the lower bound for the load rate, while the upper bound is set at 100%, which limits the load on the line to its maximum capacity. By using a step size of 1%, a set of feeder load rates can be generated.

For each load rate η_g %, combined with the net load considering the PV and load probability characteristics within supply range of substations, multiple sets of feeders can be calculated.

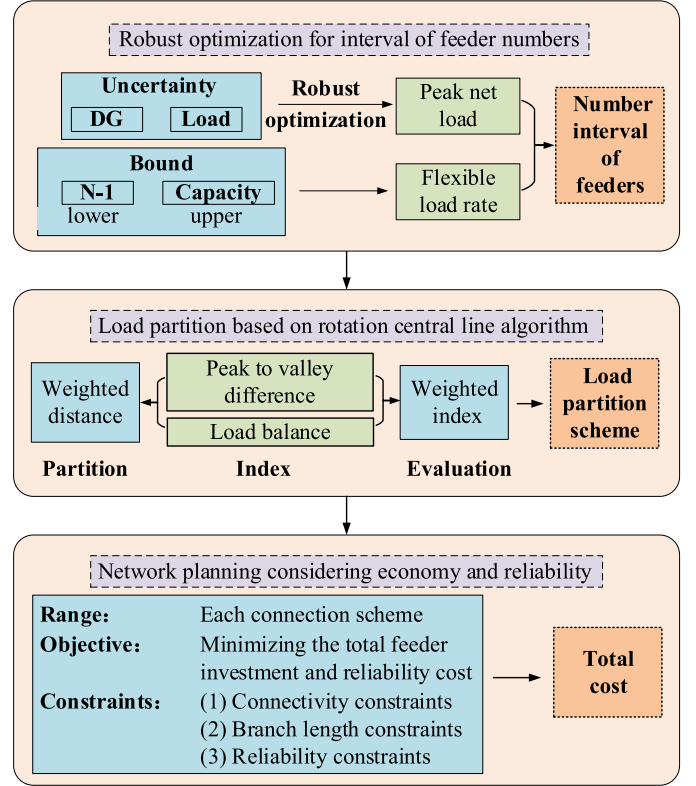


Fig. 1. The framework of planning.

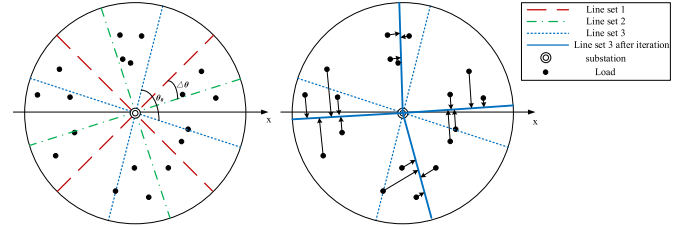


Fig. 2. Rotating central lines algorithm.

The number of feeders in group g is $N_{\text{line},g}$. The specific calculation formula is as follows. The numerator of (1) is optimized with the goal of minimizing the peak net load of substations, considering the probability characteristics and power balance of PV and load. The denominator is the relaxed capacity of a single feeder line. In order to avoid the inability of a single feeder to form a connection within the area, the total number of feeders is limited to an even number.

A. Objective Function

$$\min_{P_{\text{DR},i}^t} \max_t \frac{|\sum_{i \in N_L} P_{L,i}^t - \sum_{i \in N_{\text{PV}}} P_{\text{PV},i}^t - \sum_{i \in N_{\text{DR}}} P_{\text{DR},i}^t|}{S_{\text{line}} \cdot \eta_g \%} \quad \forall i, g \quad (1)$$

where $P_{L,i}^t$ is the conventional load demand power of i at time t , $P_{\text{PV},i}^t$ is output power of PV unit i at time t , $P_{\text{DR},i}^t$ is load demand power of i at time t , η_g % is the maximum load rate in group g ,

S_{line} is the maximum capacity of the feeder, N_{PV} , N_{L} , N_{DR} is respectively the number of PV, conventional load and shiftable load participating in demand response.

B. Constraints

1) *Power Balance Constraint*: This constraint represents the power balance constraint of PV, demand response and load, which is manifested in the following forms:

$$P_{\text{line},m}^t + \sum_{i \in N_{\text{PV},m}} P_{\text{PV},i}^t = \sum_{i \in N_{\text{L},m}} P_{\text{L},i}^t - \sum_{i \in N_{\text{DR},m}} P_{\text{DR},i}^t \quad \forall m, i, t \quad (2)$$

$$S_{\text{line}} \cdot \eta_g \% \geq (1 + \Gamma_t) P_{\text{line},m}^g \quad \forall g \quad (3)$$

where $P_{\text{line},m}^t$ is the calculated transmission power of the feeder line m , $P_{\text{PV},i}^t$ is the output power of photovoltaic unit i at time t , $P_{\text{L},i}^t$ is conventional load demand power unit i at time t , $P_{\text{DR},i}^t$ is demand response load power unit i at time t , $N_{\text{PV},m}$, $N_{\text{L},m}$, $N_{\text{DR},m}$ is respectively the number of PV, conventional load and load participating in demand response in feeder m , and Γ_t is the uncertainty parameter in this article with a value of 0.05.

2) *Uncertain Distributed Generation*: Predicting PV output is inherently complex due to uncertainties. In this study, we employ a Bayesian regression model to address this challenge [23]. Historical meteorological data and PV output are used as inputs to the Bayesian regression model to forecast PV output for the upcoming year. The same approach is applied to predict uncertain load values.

The distributed PV is modeled as a box uncertainty set $\mathbf{D}_{\text{PV}}$ as follows.

$$\begin{aligned} \mathbf{D}_{\text{PV}} = \mathbf{P}_{\text{PV}} : (1 - \Gamma_t) P_{\text{PV},\text{pre},i}^t &\leq P_{\text{PV},i}^t \\ &\leq (1 + \Gamma_t) P_{\text{PV},\text{pre},i}^t \quad \forall i \in N_{\text{PV}}, \forall t \end{aligned} \quad (4)$$

where $P_{\text{PV},\text{pre},i}^t$ is the predicted benchmark value of photovoltaic unit i at time t .

3) *Uncertain Load*: The load is modeled as a box uncertainty set $\mathbf{D}_{\text{L}}$ as follows.

$$\begin{aligned} \mathbf{D}_{\text{L}} = \mathbf{P}_{\text{L}} : (1 - \Gamma_t) P_{\text{L},\text{pre},i}^t &\leq P_{\text{L},i}^t \leq (1 + \Gamma_t) P_{\text{L},\text{pre},i}^t \\ &\quad \forall i \in N_{\text{L}}, \forall t \end{aligned} \quad (5)$$

where $P_{\text{L},\text{pre},i}^t$ is the predicted benchmark value of conventional load i at time t .

4) *Demand Response*: These constraints represent the proportion of demand response shiftable load, which is formulated as:

$$-\alpha P_{\text{L},i}^t \leq P_{\text{DR},i}^t \leq \alpha P_{\text{L},i}^t \quad \forall i, t \quad (6)$$

$$\sum_{t \in T} P_{\text{DR},i}^t = 0 \quad \forall i \quad (7)$$

where α is the maximum proportion of shiftable load to the overall load.

IV. FEEDER BLOCK PARTITION

Given the large solution space inherent in distribution network planning problems, especially when geographic information is considered, finding the optimal solution can be challenging. To address this, the planning problem is solved step by step by decoupling network planning from distribution network routing, as proposed in this study. In Section IV, the regional distribution network is divided into independent feeder blocks, thereby determining the assignment of PV units and loads.

A set of feeder number schemes within distribution network can be obtained through a robust optimization model that considers load rate relaxation. PV and load within the area can be assigned to different feeders using RCL algorithm. The resulting partitioned areas are defined as feeder blocks. The detailed division procedures are listed as follows.

A. Block Partition Considering Feeder Capacity Robustness

The net load on a feeder is reduced due to the integration of PV, allowing more traditional loads to be carried, effectively increasing load rate. However, due to the uncertainties in PV and load, it is necessary to apply robust constraints on feeder capacity to ensure reliability after faults. Furthermore, when allocating PV units and loads, the objective is to reduce the peak-valley difference in net load for each feeder and balance the net load size among feeders. To address this, the RCL algorithm is employed, using load balancing factors to represent distance. The load points in the following equations represent the net load, including PV output. The load points in the following equations represent the net load, including PV output.

1) *Robust Constraint on Feeder Line Capacity*:

$$S_{\text{line}} \cdot \eta_g \% \geq (1 + \Gamma_t) P_{\text{pline},m}^g \quad \forall g, m \quad (8)$$

2) *Weighted Distance Calculation*:

$$d_{mn}^g = \sigma_{1,m}^g \sigma_{2,mn}^g d'_{mn} \quad (9)$$

$$\sigma_{\text{lb},m}^g = \left(\frac{\sum P_{\text{pline},m}^g}{\sum P_{\text{pline},m}^g - P_{\text{pline},m}^g} \right)^k \quad (10)$$

$$\sigma_{\text{pvr},mn}^g = \left(\frac{\sum_m \varepsilon_{mn}^g}{\sum_m \varepsilon_{mn}^g - \varepsilon_{mn}^g} \right)^k \quad (11)$$

where d_{mn}^g is weighted distance from load point n to virtual central line m in group g , d'_{mn} is actual geographic vertical distance, $P_{\text{pline},m}^g$ is the peak load of block m where virtual central line m is located in group g , $\sigma_{\text{lb},m}^g$ is the weighted factor representing load balance of line m in group g , ε_{mn}^g is the difference between the peak and valley values of block m in group g , $\sigma_{\text{pvr},mn}^g$ is the weighted factor representing difference between peak and valley of line m in group g . The initial value of $\sigma_{\text{lb},m}^g$ and $\sigma_{\text{pvr},mn}^g$ is set to 1 and updated with iteration, the load rate set ranges from 50% to 100%, with a step size of 1%, and the value of η_g % in the formulas is a fixed value enumerated from the set of load rates.

Pseudo Code: Rotating central lines algorithm.

Input: $N_g, N_{\text{line}}, N_{\text{set}}, N_{\text{iter}}, P_L, P_{PV}, P_{DR}$
 $\theta_0 = 2\pi / N_{\text{line}}, \Delta\theta = \theta_0 / N_{\text{set}},$
1. **for each** load rate group **each** initial central line set
each iteration do
2. **for each** load **do**
3. calculate $P_{\text{pline},m}^g$ and ε_{mn}^g
4. calculate (d_{mn}^g) find the min d_{mn}^g load n of line m
5. $\{\Omega_{L,m,s}^g\} \leftarrow n$
6. **end for** all load
7. update $\sigma_{lb,m}^g, \sigma_{pvr,mn}^g$
8. **end for** all iteration
9. **end for** all initial central line
10. **end for** all load rate
Output: $\{\Omega_{L,m,s}^g\}$

3) Central Line Angle Positioning Method:

$$\alpha_m^g = \text{atan2} \left(\sum_{n \in \Omega_{L,m}^g} \Delta x_n, \sum_{n \in \Omega_{L,m}^g} \Delta y_n \right) \quad (12)$$

where function $\text{atan2}(x, y)$ is the angle between the radiation pointing to (x, y) and the positive direction of the x -axis, with a value range of $(-\pi, \pi)$, α_m^g is the angle of central line m in group g , (x_n, y_n) is the coordinate of load point n , and $\Omega_{L,m,s}^g$ is the set of load points belonging to line m of initial central line set s in group g .

Taking Fig. 2 as an example, it is assumed that $\eta_g \% = 50\%$, $N_{\text{line}} = 4$. There are 4 feeders in the substation, generating 3 sets of initial line as shown in the circle on the left. It can be calculated that $\theta_0 = 2\pi / N_{\text{line}} = 90^\circ$, $\Delta\theta = \theta_0 / N_{\text{set}} = 30^\circ$. In each iteration, each load is assigned to one of the third set of line according to the updated $\sigma_{lb,m}^g, \sigma_{pvr,mn}^g$. After several iterations, the final central line angle can be obtained as shown in the circle on the right.

B. Evaluation of Feeder Block Scheme

According to the aforementioned method, multiple feeder block partitioning schemes can be generated. By assigning loads with complementary characteristics to the same feeder, the peak-to-valley difference of the load supplied by each feeder can be minimized, effectively improving annual line utilization efficiency and reducing the need for large-scale line construction. To effectively mitigate the issue of high peak load stress, schemes with a lower peak-valley difference rate are selected. Additionally, to balance load distribution across similarly sized blocks, a maximum load balancing index, based on the daily maximum load, is employed to prevent uneven load distribution among blocks. In summary, this study utilizes the load balancing index and peak-to-valley difference rate to select an optimal load assignment scheme.

The better scheme of load rate group g and each initial central line set s is selected based on index A_s^g at this step. The index A_s^g is obtained by the load balance index LB_s^g and peak-to-valley

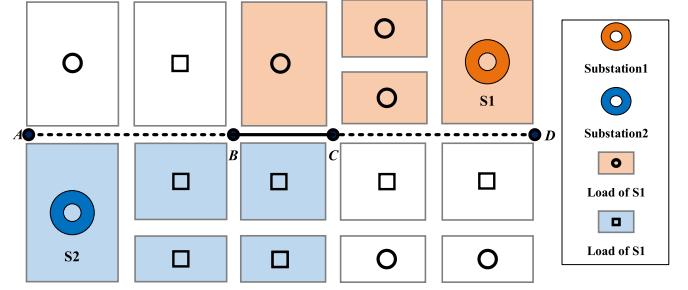


Fig. 3. Diagram of the border situation between substations.

difference rate PVR_s^g with a weighted combination.

$$A_s^g = \beta_1 LB_s^g + \beta_2 PVR_s^g \quad (13)$$

$$\beta_1 + \beta_2 = 1, \beta_1, \beta_2 \in (0, 1) \quad (14)$$

$$LB_s^g = \frac{1}{N_{\text{line}}^g} \sum_{m=1}^{N_{\text{line}}^g} \frac{L_{\text{max},m,s}^g}{L_{\text{max},s}^g} \quad (15)$$

$$PVR_s^g = \frac{1}{N_{\text{line}}^g} \sum_{m=1}^{N_{\text{line}}^g} \left(1 - \left(\max(L_{\text{max},m,s}^g) - \min(L_{\text{min},m,s}^g) \right) \right) \quad (16)$$

where N_{line}^g is the number of feeders in load rate group g , β_1 and β_2 are weighting factors. $L_{\text{max},s}^g$ is the maximum load of all feeders in load rate group g and central line set s , $L_{\text{max},m,s}^g$ is the maximum load of feeder m in load rate group g and central line set s , and $\Omega_{L,s}^g$ is a set of all loads in group g and central line set s .

V. NETWORK PLANNING CONSIDERING ECONOMY AND RELIABILITY

There are more potential positions for tie switches in feeder blocks with high border connectivity, providing greater optimization flexibility during the routing stage. It is evident that feeder blocks within the same substation are all adjacent to each other. Therefore, greater attention is given to feeder blocks adjacent to those in different substations. The degree of adjacency between feeder blocks is denoted by the index B_{com} .

$$B_{\text{com}} = \sum_{i=1}^{N_{\text{com}}} b_i^{\text{bor}} \quad \forall i \in N_{\text{com}} \quad (17)$$

$$b_i^{\text{bor}} = \frac{b_{i,1} \cap b_{i,2}}{b_{i,1} \cup b_{i,2}} \quad \forall i \in N_{\text{com}} \quad (18)$$

where N_{com} is the number of combined feeder blocks between different substations. b_i^{bor} is the degree of overlap between the boundaries of the two feeder blocks. Power supply units divided between substations, $b_{i,1}$ and $b_{i,2}$ is the adjacent boundary lengths of feeder blocks 1 and 2 within the combined block.

For example, as illustrated in Fig. 3, B and C are optional positions for tie switch, $b_i^{\text{bor}} = (AC \cap BD) / (AC \cup BD) = BC$. By summing up the degree of adjacency b_i^{bor} for each feeder

division scheme, the schemes with the highest connectivity can be selected.

A. Objective Function

The objective function primarily focuses on costs associated with main lines, branch lines, and reliability expenses, acknowledging the trade-off between improving reliability and the investment costs of the lines. To achieve overall economic optimality, a balance between these factors must be sought.

Based on feeder block combination schemes, the routing of main and branch lines will be carried out separately, with the goal of minimizing total investment and operational costs. However, due to the relaxation of the load rate, there may be a decrease in reliability for certain loads. To appropriately configure the placement of sectionalizing and tie switches, it is necessary to calculate the reliability level of the feeder line during routing.

Each feeder in the interconnection is decoupled from the shortest length of the main and branch feeders during routing, ensuring that only the power margin of the connecting feeders affects each other. Therefore, the reliability and economic cost calculation for each feeder line is as follows:

$$\min C_h = \sum_m (C_{h,m}^{\text{tr}} + C_{h,m}^{\text{br}} + C_{h,m}^{\text{re}}) \quad \forall h \in N_{\text{com}} \quad (19)$$

where C_h is the total cost of each pair of interconnected feeders in group h , $C_{h,m}^{\text{tr}}$ and $C_{h,m}^{\text{br}}$ represent the investment and operation costs for the main and branch feeders of the single-ring network of feeder m in group h , $C_{h,m}^{\text{re}}$ is the annual power outage cost of feeder m in group h , and N_{com} is the number of interconnected feeder groups combined by feeder blocks.

$$C_{h,m}^{\text{trk}} = \frac{f(1+f)^r}{(1+f)^r - 1} (1+\alpha) \sum_{(i,j)} \chi_{ij} \text{len}_{ij} c_{\text{trk}}(i,j) \in \{\Omega_h^{\text{str}}\} \quad (20)$$

where f is the discount rate, r is the service life of line, α is a ratio of operation and maintenance cost, χ_{ij} is a binary variable representing whether to construct a line between node i and j , c_{trk} is the construction cost per unit length of main feeder, len_{ij} is the length of feeder constructed between i and j , and Ω_h^{str} is a set of endpoints of all pending street segments.

$$C_{h,m}^{\text{brh}} = \frac{f(1+f)^r}{(1+f)^r - 1} (1+\alpha) \sum \text{len}_n c_{\text{brh}} n \in \Omega_h^{\text{load}} \quad (21)$$

where len_n is the length of feeder constructed between load n and main line, and c_{brh} is the construction cost per unit length of branch feeder.

$$C_{h,m}^{\text{re}} = F_n \cdot \sum_{(i,j)} \lambda \cdot \text{len}_{ij} L_n t_{ij,n}(i,j) \in \{\Omega_h^{\text{str}}\} \quad (22)$$

where F_n is reliability loss compensation cost coefficient, which is related to the type of load. λ is the fault rate of all feeder sections, L_n is the load value of load point n , and $t_{ij,n}$ is the power outage time of load point n after the fault of feeder section ij .

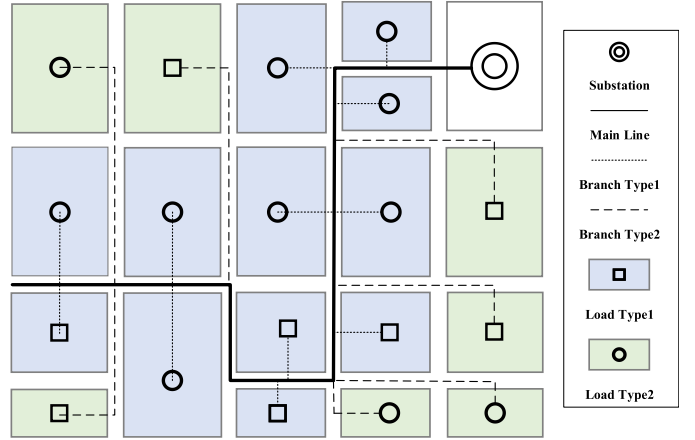


Fig. 4. Connection method of Branch lines.

B. Constraints

1) Connectivity Constraints:

$$\sum \chi_{sj} = 1 \quad j \in \{\Omega_h^{\text{V}}\} \quad s \in \{\Omega_h^{\text{Sub}}\} \quad (23)$$

$$\sum \chi_{ie} = 1 \quad i \in \{\Omega_h^{\text{V}}\} \quad e \in \{\Omega_h^{\text{Tie}}\} \quad (24)$$

where Ω_h^{V} is a set of all nodes in group h , Ω_h^{Sub} is a set of all substation nodes in group h , and Ω_h^{Tie} is a set of all tie switch nodes to be selected in group h .

A virtual network is established to achieve open-loop connectivity of the topology structure through the following constraints. It is assumed that the substation node is the source node, and the virtual load at each node is set to 1.

$$\sum_{j \in \Omega_i^{\text{child}}} F_{ij} - \sum_{k \in \Omega_i^{\text{parent}}} F_{ki} = -1 \quad \forall i \in \{\Omega_h^{\text{V}} \setminus \Omega_h^{\text{Sub}}\} \quad (25)$$

$$-\chi_{ij} (N_h^{\text{V}} - N_h^{\text{Sub}}) \leq F_{ij} \leq \chi_{ij} (N_h^{\text{V}} - N_h^{\text{Sub}}) \quad \forall ij \in \Omega_h^{\text{str}} \quad (26)$$

$$0 \leq \sum_{j \in \Omega_i^{\text{child}}} F_{ij} \leq (N_h^{\text{V}} - N_h^{\text{Sub}}) \quad \forall i \in \{\Omega_h^{\text{Sub}}\} \quad (27)$$

where Ω_i^{child} and Ω_i^{parent} are the sets of child and parent nodes for node i , N_h^{V} is the number of nodes in the constructed feeder in combination group h , N_h^{Sub} is the number of substations in the constructed feeder in combination group h , and F_{ij} is virtual power flow from node i to node j .

2) Branch Length Constraint: As illustrated in Fig. 4, the load can be categorized into two types. For load type 1, at least one section of the main line is constructed along streets surrounding the load. The branch line length for load type 1 can be directly calculated as the shortest distance between the load and the street where the line section is constructed. For load type 2, where there is no main line section nearby, it is necessary to identify the nearest street for connection. The branch line length for load type 2 can be simplified to the shortest distance between the load and any node of the main line. Given the regular shape of the street layout, the difference in geographical coordinates between the load and the street node can be used to represent

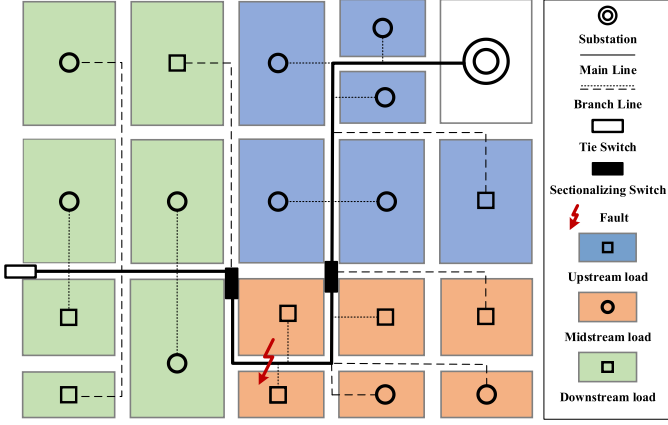


Fig. 5. Analysis of the impact of power outage time.

this shortest distance. The specific expression is as follows:

$$z_{n,ij} \leq \chi_{ij} \quad \forall n, ij \quad (28)$$

$$\sum_{ij} z_{n,ij} \leq 1 \quad \forall ij \in G_h^n \quad \forall n \in \Omega_h^{\text{load}} \quad (29)$$

$$\text{len}_n \leq D_{n,ij} + (1 - z_{n,ij}) \cdot M \quad \forall ij \in \Omega_{E,n}^h \quad \forall n \in N_{\text{load}}^h \quad (30)$$

where $z_{n,ij}$ is a binary variable representing whether streets around the load are constructed, G_h^n is a set of streets surrounding the load n , and $D_{n,ij}$ is the distance from load node n to street segment ij .

$$\kappa_v \geq \chi_{ij} \quad ij \in \{\Omega_h^{\text{str}}\} \quad v \in \{i, j\} \quad (31)$$

$$\Delta x_{n,v} \geq x_n - x_v - (\kappa_v - 1) \cdot M \quad \forall v \in \Omega_h^V \quad \forall n \in \Omega_h^{\text{load}} \quad (32)$$

$$\Delta x_{n,v} \geq x_v - x_n - (\kappa_v - 1) \cdot M \quad \forall v \in \Omega_h^V \quad \forall n \in \Omega_h^{\text{load}} \quad (33)$$

$$\Delta y_{n,v} \geq y_n - y_v - (\kappa_v - 1) \cdot M \quad \forall v \in \Omega_h^V \quad \forall n \in \Omega_h^{\text{load}} \quad (34)$$

$$\Delta y_{n,v} \geq y_v - y_n - (\kappa_v - 1) \cdot M \quad \forall v \in \Omega_h^V \quad \forall n \in \Omega_h^{\text{load}} \quad (35)$$

$$\text{len}_n \leq \Delta x_{n,v} + \Delta y_{n,v} + M \cdot (1 - \phi_{n,v}) \quad \forall v \in \Omega_h^V \quad \forall n \in \Omega_h^{\text{load}} \quad (36)$$

$$\sum \phi_{n,v} \leq 1 \quad \forall v \in \Omega_h^V \quad (37)$$

where (x_n, y_n) is the coordinate of the load n , $(\Delta x_{n,v}, \Delta y_{n,v})$ are the differences in horizontal and vertical coordinates between load n and node v , $\phi_{n,v}$ is a binary variable representing whether the distance between node v and n is minimum value of load n . κ_v is a binary variable representing whether the node v is included in constructed feeders. Ω_h^V is a set of four vertices of each load point in group h , and Ω_h^{load} is a set of all loads in group h .

3) *Reliability Constraints*: As shown in the Fig. 5, each feeder is equipped with two segmented switches, which are connected to the other feeder through a tie switch. All loads are divided into three sections. Within the same section, the duration

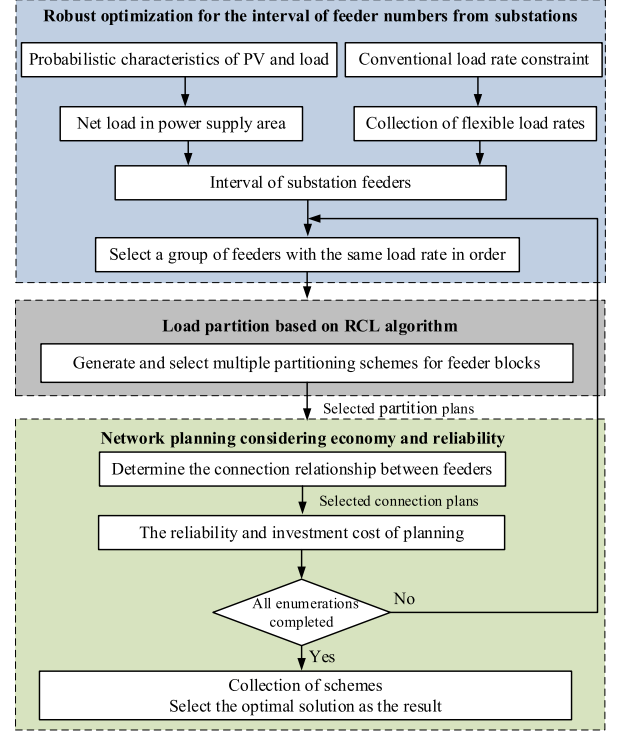


Fig. 6. The procedure of solution.

of a power outage is identical. For instance, in the event of a fault in a midstream line section, the outage time for upstream loads corresponds to the fault isolation time, while the midstream outage time is equal to the fault restoration time. The outage time for downstream loads depends on the transfer capacity of the tie line. If these loads can be transferred by the connected feeder, their outage time will be the fault isolation and transfer time; otherwise, it will be the fault repair time.

Based on the above analysis, the following constraints can be applied to the outage time of load point n after a fault occurs on branch ij :

$$t_{ij,n} \geq t_{\text{ST}} \quad (38)$$

$$t_{ij,n} \geq \varsigma_{ij,n} t_{\text{SRT}} \quad (39)$$

$$t_{ij,n} \geq \tau_{ij,n} t_{\text{TTR}} \quad (40)$$

where $\varsigma_{ij,n}$ is a binary variable indicating whether load n can be transferred, and $\tau_{ij,n}$ is a binary variable representing whether the load n can be restored only after the fault is repaired.

a) *Order variable constraint*:

$$q_i + 1 \leq q_j + (1 - \chi_{ij}) \cdot M \quad \forall (i, j) \in \Omega_h^{\text{str}} \quad (41)$$

$$q_i + 1 \leq q_j + (1 - \chi_{ji}) \cdot M \quad \forall (i, j) \in \Omega_h^{\text{str}} \quad (42)$$

where q_i is a sequential continuous variable that represents the order relationship of each node in the main trunk line.

$$q_j \leq q_{s1} + (1 - \delta_{ij}^U) \cdot M \quad \forall (i, j) \in \Omega_h^{\text{str}} \quad (43)$$

$$q_i \leq q_{s1} + (1 - \delta_{ji}^U) \cdot M \quad \forall (i, j) \in \Omega_h^{\text{str}} \quad (44)$$

$$q_{s1} \leq q_i + (1 - \delta_{ij}^M) \cdot M \quad \forall (i, j) \in \Omega_h^{\text{str}} \quad (45)$$

$$q_j \leq q_{s2} + (1 - \delta_{ij}^M) \cdot M \quad \forall (i, j) \in \Omega_h^{\text{str}} \quad (46)$$

$$q_{s1} \leq q_j + (1 - \delta_{ji}^M) \cdot M \quad \forall (i, j) \in \Omega_h^{\text{str}} \quad (47)$$

$$q_i \leq q_{s2} + (1 - \delta_{ji}^M) \cdot M \quad \forall (i, j) \in \Omega_h^{\text{str}} \quad (48)$$

$$q_{s2} \leq q_i + (1 - \delta_{ij}^D) \cdot M \quad \forall (i, j) \in \Omega_h^{\text{str}} \quad (49)$$

$$q_{s2} \leq q_j + (1 - \delta_{ji}^D) \cdot M \quad \forall (i, j) \in \Omega_h^{\text{str}} \quad (50)$$

$$\delta_{ij}^U + \delta_{ij}^M + \delta_{ij}^D \leq \chi_{ij} \quad \forall (i, j) \in \Omega_h^{\text{str}} \quad (51)$$

$$\delta_{ji}^U + \delta_{ji}^M + \delta_{ji}^D \leq \chi_{ji} \quad \forall (i, j) \in \Omega_h^{\text{str}} \quad (52)$$

$$q_v \leq q_{s1} + (1 - \delta_v^U) \cdot M \quad \forall v \in \Omega_h^V \quad (53)$$

$$q_{s1} \leq q_v + (1 - \delta_v^M) \cdot M \quad \forall v \in \Omega_h^V \quad (54)$$

$$q_v \geq q_{s2} + (1 - \delta_v^D) \cdot M \quad \forall v \in \Omega_h^V \quad (55)$$

$$\delta_v^U + \delta_v^M + \delta_v^D \leq \kappa_v \quad \forall v \in \Omega_h^V \quad (56)$$

where $\delta_{ij}^U, \delta_{ij}^M, \delta_{ij}^D, \delta_v^U, \delta_v^M, \delta_v^D$ are binary variables corresponding to the upstream, midstream, and downstream segments for branch ij and node v respectively, and q_{s1}, q_{s2} are order variables representing the positions of segment switches $s1$ and $s2$.

$$L_{ij}^U \geq L_n \cdot z_{n,ij} + L_n \cdot \phi_{n,i} + L_n \cdot \phi_{n,j} - M \cdot (1 - \delta_{ij}^U) \quad (57)$$

$$L_{ij}^U \geq 0 \quad (58)$$

Using the upstream branch segment information as an example, L_{ij}^U is the load value of upstream branch ij , and L_n is the load value of load point n . This analysis considers failures in the main trunk line.

b) Constraint of sectionalizing switch position:

$$W_1 = \sum_{(i,j)} L_{ij}^U \quad \forall (i, j) \in \Omega_h^{\text{str}} \quad (59)$$

$$W_2 = \sum_{(i,j)} L_{ij}^M \quad \forall (i, j) \in \Omega_h^{\text{str}} \quad (60)$$

$$W_3 = \sum_{(i,j)} L_{ij}^D \quad \forall (i, j) \in \Omega_h^{\text{str}} \quad (61)$$

$$W_1 - W_2 \leq \Delta W_1 \quad W_2 - W_1 \leq \Delta W_1 \quad (62)$$

$$W_1 - W_3 \leq \Delta W_2 \quad W_3 - W_1 \leq \Delta W_2 \quad (63)$$

$$W_3 - W_2 \leq \Delta W_3 \quad W_2 - W_3 \leq \Delta W_3 \quad (64)$$

$$\Delta W_i \leq \zeta i \in \{1, 2, 3\} \quad (65)$$

(59)–(65) are designed to constrain the load balancing across each segment. W_1, W_2 and W_3 are the sums of the loads for all branches in each section, $\Delta W_1, \Delta W_2$ and ΔW_3 are all intermediate variables, and ζ is a constant that restricts the load difference across each section.

c) Constraint of sectionalizing switch position:

$$\delta_{ij}^W = 3\delta_{ij}^U + 2\delta_{ij}^M + \delta_{ij}^D - 3\delta_n^U - 2\delta_n^M - \delta_n^D \quad (66)$$

$$\delta_{ij}^W - M(1 - \varsigma_{ij,n}) \leq \varsigma_{ij,n} \leq \delta_{ij}^W \quad (67)$$

$$S_{\text{Tie}} = S_{\text{line}} \cdot (1 - \eta_g \%) \quad (68)$$

$$S_{\text{Tie}} - Mv_{st} \leq W_{\text{SRT}} \leq S_{\text{Tie}} - 1/M - M(1 - v_{st}) \quad (69)$$

$$W_{\text{SRT}} \geq L^M + L^D - M(1 - \delta_n^M) \quad (70)$$

$$W_{\text{SRT}} \geq L^D - M(1 - \delta_n^D) \quad (71)$$

$$(1 - v_{st}) \leq m_1 \leq (1 - v_{st})M \quad (72)$$

$$\delta_{ij}^W - M(1 - m_2) \leq m_2 \leq \delta_{ij}^W \quad (73)$$

$$(m_1 + m_2)/2 \leq \tau_{ij,n} \leq m_1 + m_2 \quad (74)$$

where δ_{ij}^W is an intermediate variable constructed to represent the positional relationship between the faulted branch ij and the load point n , S_{Tie} is the margin of the connecting feeder line, W_{SRT} representing the total amount of loads that need to be switched, v_{st} is a binary variable indicating whether the load can be transferred, with a value of 1 indicating that it can be switched. m_1 is a binary variable that is set to 1 when the load cannot be transferred, m_1 is set to 1. When the faulty line is in the same section as the load, it is necessary to wait for the fault to be repaired before resuming power supply.

To ensure a minimum level of reliability in the constraints, the reliability index ASAI is calculated as follows:

$$ASAI = 1 - \frac{\sum_n \lambda \cdot \text{len}_{ij} t_{ij,n} N_n}{8760 \sum_n N_n} \geq ASAI_{\text{need}} \quad (75)$$

where N_i is the total number of consumers, and $ASAI_{\text{need}}$ is the lowest bound of ASAI index.

VI. THE PROCEDURE OF PLANNING

The specific solution strategy for regional distribution network planning considering the probability characteristics of PV and load under differentiated reliability requirements is shown in Fig. 6.

A. Robust Optimization for the Interval of Feeder Numbers From Substations

Step1: Calculate the peak net load within the substations' service range based on the probabilistic characteristics of PV and load uncertainty.

Step2: Derive multiple feeder intervals using flexible load rates. Based on the upper and lower bounds of the load rates and specified increments, generate a collection of flexible load rate sets.

Step3: Incorporate the peak net load values obtained in Step 1 to determine various scenarios for the number of outgoing feeders. Sequentially select a set of feeder numbers corresponding to each flexible load rate.

B. Load Partition Based on RCL Algorithm

Step4: Implement the RCL algorithm to generate multiple partitioning schemes for feeder blocks. Subsequently, evaluate and select several schemes based on weighted indicators.



Fig. 7. Street geographic information and load location map.

C. Network Planning Considering Economy and Reliability

Step5: For each load assignment scheme, generate combinations of feeder connections based on the degree of adjacency between feeder blocks.

Step6: Plan the routing of each group of connected feeders individually to achieve a regional network planning scheme that minimizes both reliability costs and the costs associated with main trunk and branch lines.

Step7: Verify if all possible enumerations have been examined. If not, return to step 3.

Step8: Compare the network planning schemes across various flexible feeder load rates and select the scheme with the lowest total cost as the optimal planning solution.

VII. CASE ANALYSIS

The region is equipped with three 2×31.5 MVA substations, connected to 80, 83, and 80 loads respectively. The feeder capacity is 7.8 MW. There are four types of loads: residential, commercial, industrial, and administrative. A total of 152 street nodes and 265 street sections fall within the power supply range of the three substations. PVs are connected to existing load points. The geographical locations of each street node, street section, and load are shown in Fig. 7 and Table II.

The main line costs 100000 yuan/km, while the branch line costs 70000 yuan/km. The fault frequency of the feeder section is 0.047 times/(year·km), with a fault repair time of 3.73 hours, fault isolation time of 0.5 hours, and transfer operation time of 1 hour. Simulations were conducted using CPLEX on a laptop with a 12th Gen Intel Core™ i7-12700H CPU at 2.30 GHz and 16GB RAM.

A. Robust Optimization Model for Feeder Numbers From Substations:

The number of feeders from substations is shown in Table III. According to the N-1 criterion, the lower bound of the load rate is 50%, and the upper bound can reach 100% based on line

TABLE II
NUMERICAL VALUES USED FOR SIMULATION

Features	Values
Unit main line cost	100000 (yuan/km)
Unit branch line cost	70000 (yuan/km)
Residential load compensation	10.175(yuan/kWh)
Commercial load compensation	95.643(yuan/kWh)
Industrial load compensation	176.928(yuan/kWh)
Administrative load compensation	21.218(yuan/kWh)
Rate of discount f	0.08
Service life of line r	20 years
Fault repair time	3.73 hours
Transfer operation time	1 hour
Fault isolation time	0.5 hour
Capacity of feeder	7.8 MW

TABLE III
THE NUMBER OF FEEDERS FROM SUBSTATIONS

Load Rate	Number of feeders			
	S1	S2	S3	Sum
0.5	10	9	9	28
0.51~0.54	9	9	8	26
0.55~0.59	8	8	8	24
0.6~0.67	8	7	7	22
0.68~0.7	7	7	6	20

TABLE IV
COMPREHENSIVE COST OF PLANNING SCHEME

Number of feeders	Total investment of feeders (10000 yuan)	Reliability cost (10000 yuan)	Comprehensive cost(10000 yuan)
28	156.3099	230.0236	386.3329
26	142.3136	235.7732	378.0868
24	140.8258	236.3879	377.2137
22	143.2348	277.3843	420.6191
20	140.6064	281.2718	421.8782

TABLE V
COMPARISON OF SOLUTION METHODS

Model	Reliability method	Solution time(s)
Ant colony	Monte Carlo	569.462318
MILP	Analytical	387.931230

capacity. However, in practice, when the load rate exceeds 70%, the power transfer capacity of the line after a fault is severely limited, resulting in poor reliability. Therefore, the upper bound of the load rate is chosen as 70% in this study. As the load rate is relaxed, the number of feeders from substations decreases.

The seasonal curves for PV, residential, commercial, industrial, and administrative loads are shown in Fig. 8, corresponding to the four seasons: spring, summer, autumn, and winter. Because distributed photovoltaic output can partially meet load demand, the overall net load peak is reduced, alleviating the power supply pressure on the medium voltage feeder. At the 50% load rate level, the number of feeders from the three substations

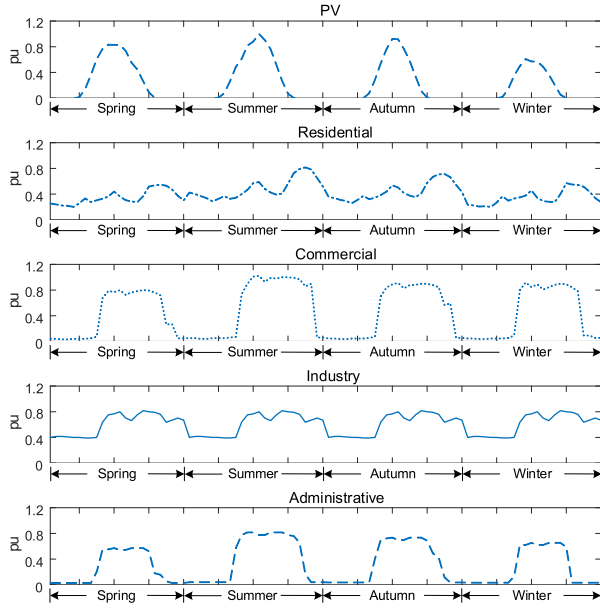


Fig. 8. Sequential output curves of PV and load.

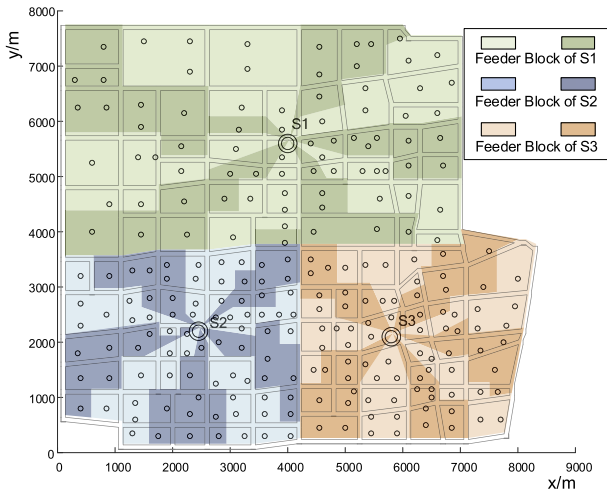


Fig. 9. Load partition without PV.

is 10, 10, and 10, respectively, without PV. After integrating PV, the number of feeders decreases to 10, 9, and 9, respectively, due to the reduction in net load peak within the substations' power supply range, as shown in Figs. 9 and 10.

B. Robust Optimization Model of Feeders

There are 10 feeders from substations under rigid N-1 constraints. Fig. 11 illustrates detailed planning schemes for two sets of connected feeders in substations S2 and S3. The colored areas indicate the formation of four connected feeders with high adjacency. Under the relaxed load rate interval $[0.55 \sim 0.59]$, there are 8 feeders from substations. Two sets of connected feeders between substations S2 and S3 were also selected for comparison. The local routing results of the optimal planning scheme after load rate relaxation are shown in Fig. 12.

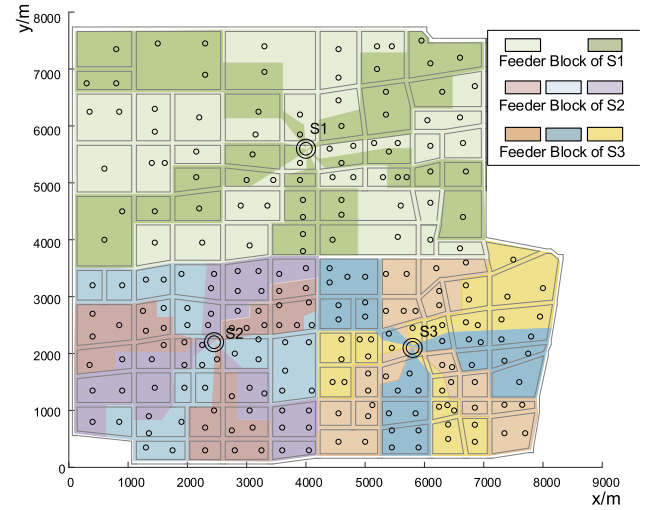


Fig. 10. Load partition integrated with PV.

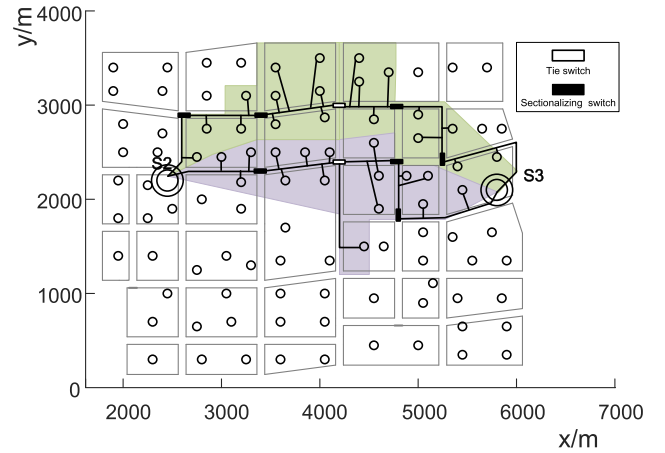


Fig. 11. Detailed planning scheme under rigid N-1.

Both figures display the routing mode of the main trunk line and branch line. The RCL algorithm effectively avoids feeder crossings. Moreover, it is evident that feeders with a relaxed load rate can carry more load, fully utilizing the feeders' capacity. The tie switches in Figs. 11 and 12 are configured at the borders of feeder blocks to leverage geographical features. Based on the rigid N-1 criterion, with sufficient feeders, the length of each feeder is too short to place two sectionalizing switches, allowing for a reduction in the number of switches. However, after relaxing the load rate, the position of each sectionalizing switch becomes critical for fault isolation and load transfer. To effectively improve system reliability, switches must be reasonably distributed, ensuring that the loads carried by each section are approximately equal.

This paper examines 5 groups of feeders from 3 substations, retaining 8 load partition schemes and 2 combination schemes, totaling 80 schemes to compare and select from. Cases 1-16 represent the schemes under the first group's feeder count with a 50% load rate. Cases 16-32 cover the second group with a load rate of 51%-54%, and so forth. The cost distribution for main

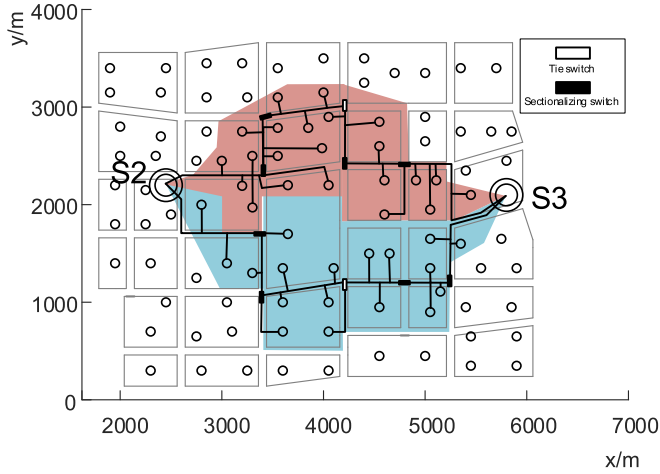


Fig. 12. Detailed planning scheme under relaxed load rate.

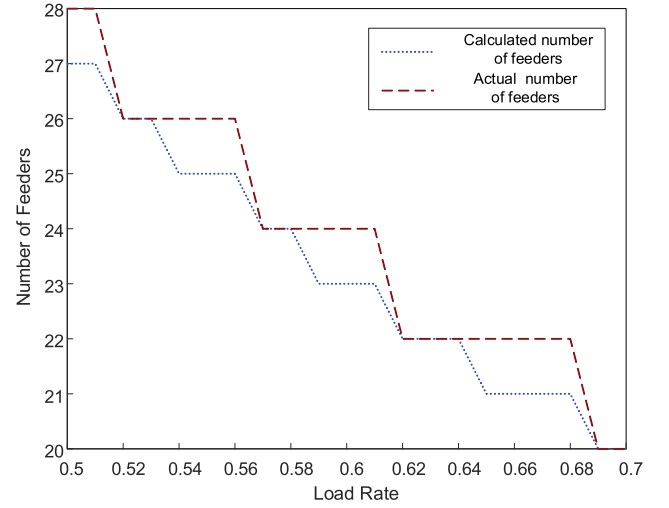


Fig. 14. Relationship between feeder number and load rate.

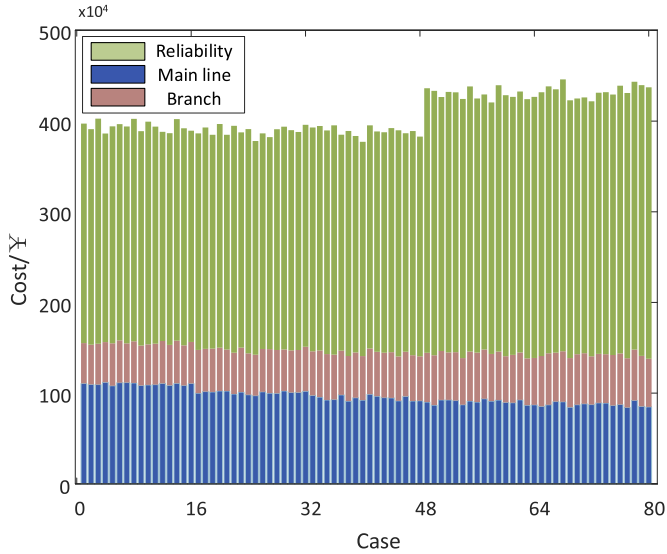


Fig. 13. Total cost of test cases.

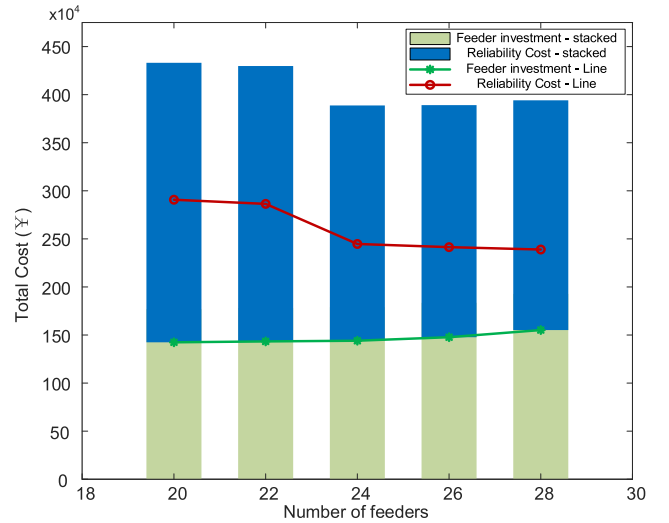


Fig. 15. Relationship between feeder number and cost.

trunks, branch lines, and reliability is illustrated in the stacked bar chart.

In Fig. 13, as the number of feeders decreases, the investment in the main feeder shows a gradual horizontal decrease, while the investment in branch feeders increases. Meanwhile, the reliability cost exhibits a gradual upward trend, with a significant spike observed in the fourth group of load rate relaxation intervals. Due to the decreasing number of feeders, the load carried by each feeder increases to a certain extent, leading to insufficient transfer margins after a fault, which in turn causes a sharp increase in reliability costs. If reliability costs are not considered, it's easy to devise a planning scheme that merely reduces the length of main trunk feeders, sacrificing reliability levels to cut feeder costs. Therefore, incorporating a reliability-based annual outage loss fee into the model helps to achieve a more reliable and economical planning scheme.

Table IV presents the specific values for the optimal route investment cost, reliability cost, and comprehensive cost for each group of feeders. The optimal power supply unit division scheme

is found in Case 40, with 24 feeders in the load rate interval of $[0.55 \sim 0.59]$, which balances both investment and minimization of power outage losses.

C. Sensitivity Analysis

As shown in Fig. 14, as the load rate gradually increases from 0.5 to 0.7, the number of feeders from substations decreases from 28 to 20. The red dashed line represents the actual number of feeders, adhering to the single tie-line rule. The blue dotted line represents the total number of feeders based on the increased load rate without considering tie-line connections, which may result in radial lines that cannot form tie connections. It is evident that as the load rate is relaxed, the number of feeders from substations significantly decreases, which is crucial for reducing line investment.

The average reliability cost and line investment cost for the 16 sets of schemes corresponding to each number of feeders are plotted as two curves in Fig. 15. The red curve represents

the average reliability cost, which increases as the number of feeders decreases. The green curve represents the average feeder investment cost, which decreases as the number of feeders decreases. It can be observed that when the number of lines decreases to 22, the reliability level drops, leading to higher reliability costs. However, when the number of feeders is 24, the total cost of the planning scheme is minimized.

D. Comparison of Efficiency

The planning based on traditional heuristic Ant Colony algorithms and Monte Carlo reliability evaluations was also applied to the test system. Although different reliability assessment methods were used, the reliability calculations required some time due to the distribution system's size. The method proposed in this paper shows advantages in solving time, as it can divide the entire region into several feeder blocks, which is more conducive to efficient problem-solving shown in Table V.

VII. CONCLUSION

In this article, a regional distribution network planning method is proposed considering differentiated reliability demand and probability characteristics of PV and load. Firstly, the number interval of total feeders from substations is optimized on basis of the robust net load value and relaxed load rate. Then, aiming at balancing the net load between feeders and minimizing peak-valley differences, a load partition method is proposed based on RCL algorithm. Finally, a systematic planning model considering reliability and economy is proposed in service area of each feeder. The following conclusions can be drawn:

- 1) The method optimizes the number interval of total feeders from substations by considering flexible load rates and the probabilistic characteristics of PV and load. Consequently, power supply range of each feeder can be divided by leveraging flexible capacity constraints.
- 2) Compared with traditional methods, the proposed method effectively handles the complex relationship among the total number of feeders from substations, investment costs, and reliability costs. It can be obtained the optimal planning scheme by coupling reliability and economy.

Further research will focus on planning methods for expanding developed areas by integrating a high proportion of distributed energy resources with high reliability demands.

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